

w6	3/31	AI Fluency	<i>“AI Fluency: Framework & Foundations”</i> by Anthropic
	4/02	Agentic AI Coding	Workflow with VS Code, Jupyter Notebook, GitHub
	4/07	Midterm Review	
w7	4/09	Midterm Exam	
	4/11	Research Workshop II	Proposal refinement, hypothesis, measurement
<i>Assignment 2 (revised proposal: data & empirical strategy) is due on 4/12 at 23:59.</i>			

Midterm Exam

(No make up exam)

- Close-book, one A4 cheat sheet (both sides, handwritten or printed)
- Written exam only, no computer/ smartwatch / translation pen
- 30 single- and multiple-choice questions (30% of total course grade)
- Test ability to read and understand Python code, identify and debug errors



温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3001 WHS01
Python Programming for Business

AI Fluency

Dawei Chen

Week 6-1, 31 March 2026

Learning Objective

By the end of this session, you should be able to:

- Have a broader view of AI use than prompt engineering
- Understand and apply [4D Fluency Framework](#)
 - **Delegation** — Decompose a task and decide which parts to assign to AI and which to own yourself
 - **Description** — Write clear, scoped prompts that give AI enough context to generate useful content
 - **Discernment** — Read, test, and critically evaluate AI-generated content rather than accepting it at face value
 - **Diligence** — Apply responsible and transparent practices when using AI assistance in your work

Opening

1) How do you use AI in daily life?

What tasks do you use AI for? Study? Writing? Translation? Planning? Coding? Search?

2) How do you usually interact with AI?

- **Passive:** ask one question → accept the answer
- **Engaged:** ask for a summary, explanation, or ideas → read it over
- **Iterative:** ask → review → refine the prompt → ask again
- **Delegating:** break a task into steps → direct AI through each one → verify the result

Translate this

Reply this email

Write me a research proposal
about AI's impact on business

Write Python code to
collect data in XHS

Shift to Python

Q1

Have you used AI for coding before?

Q2

What kinds of tasks did you use it for?

Q3

What worked well?

Q4

What went wrong – or felt off?

AI helps here	You still own this
Generate starter code	Define the right problem
Write comments	Know if the output is correct
Explain error messages	Debug logic errors
Explain unfamiliar code	Connect code to your data

What Is AI Fluency

What it is not

Memorize prompt templates

Learn platform-specific shortcut

Get better output by trial and error

A skill that expires when tool change

What it is

A mindset – how you think about AI

A collaboration skill – how you think with AI

A grounding framework for human-AI work

A transferable capability that lasts

AI fluency means using AI in ways that are:



Effective

Get the right outcome



Efficient

Use times & effort well



Ethical

Keep humans in the loop



Safe

Avoid harm & misuse

“Prompt engineering may change quickly — AI fluency is the more stable capability.”

Three Ways to Collaborate with AI

MODE 1

Automation

You instruct. AI executes.

BEST FOR

- Repetitive or formulaic tasks
- Format, summarize, clean
- First drafts and starter code

USE WHEN

- Goal is clear and narrow
- Output is easy to verify

MODE 2

Augmentation

You and AI think together.

BEST FOR

- Brainstorm & problem solve
- Refine ideas or approaches
- Debug & improve code

USE WHEN

- Task needs iteration & judgment
- You want feedback or alternatives

MODE 3

Agency

You configure. AI acts.

BEST FOR

- Multi-step or recurring workflows
- Systems that act on your behalf
- Scheduled or automated pipelines

USE WHEN

- Process is well-defined with guardrails
- You are ready to monitor outcomes

Limitations of Generative AI

Hallucination

AI context window

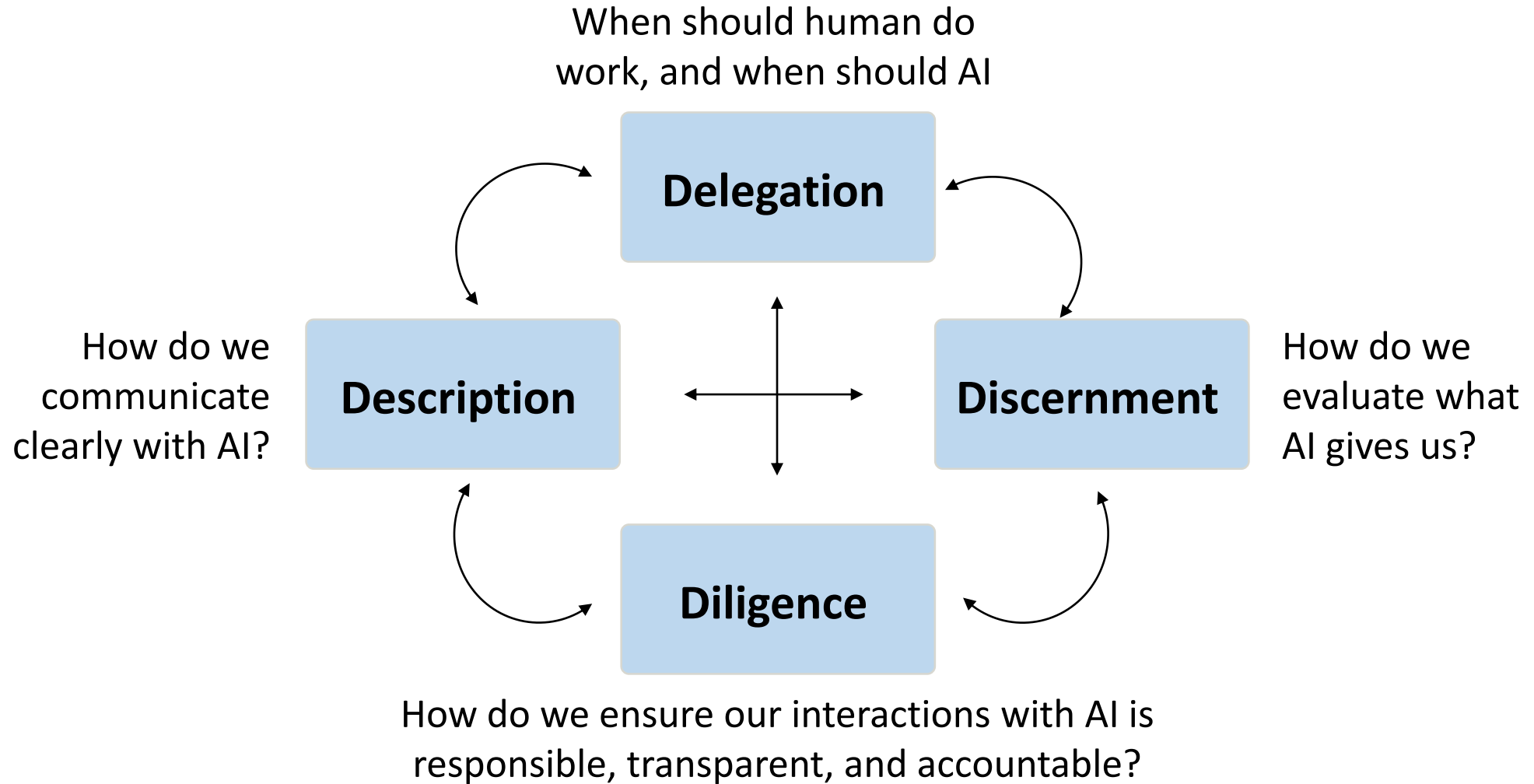
Reasoning limitation

Knowledge cutoff date

Non-deterministic output

Humans provide	AI provides
Critical thinking	Speed
Judgement	Scale
Creativity	Pattern recognition
Ethical oversight	Processing abilities

AI Fluency Framework



Delegation

Understanding the problem you're trying to solve

Understanding the capabilities of available AI tools

Breaking down complex work into smaller parts

Making strategic decisions about who does what

Choosing the right mode of interaction

Delegation

Problem Awareness

The ability to clearly define your goals and understand what work is needed before involving AI tools

- What does “success” look like
- What kind of thinking and work is needed to get there?
- AI fluency begins with and depends on your expertise

Platform Awareness

A working knowledge of available AI systems and their specific capabilities and limitations

- Understand the unique strengths and limitations of available options
- Which models perform best for your task?
- Which system prioritize speed, or creativity, or depth, or accuracy

Task Delegation

The strategic process of dividing work between humans and AI

- What could be usefully automated?
- Where would augmentation create more values than working separately?
- What would be done by a human alone?
- What could be done by an agent on your behalf?

Description

Don't just craft prompts

Explain tasks, ask questions, provide context, and guide the interaction

Build a shared thinking environment where both you and the AI can each do your best work

Description

Product Description

The ability to clearly define the characteristics of your desired output

- Clear describe what you want
- Context, format, audience, style, and other constraints
- Give AI the information it needs to deliver what you're actually looking for, not just assuming what you want

Process Description

The ability to guide the AI's thought process

- "How" can be more important than "What"
- Providing training specific to your problem
- Specific data, key tasks, preferred order, and so on

Performance Description

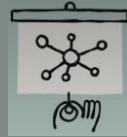
The ability to define the behavioral aspects of an AI interaction

- AI tools are interactive systems that can behave differently in different contexts
- You need to explain how you want the AI to behave to get the best results

Effective Prompting Techniques



Provide context



Offer examples



**Specify output
constraints**



**Break down
complex tasks**



**Give the AI space
to think**



Define roles

Discernment

Your ability to critically evaluate what AI produces, how it produces it and how it behaves

You need

Domain expertise

An understanding of how AI systems work and their typical shortcomings

Discernment

Product Discernment

Your capacity to judge the quality of the AI output

- Factually accurate?
- Appropriate to audience and purpose?
- Coherent and well-structured?
- Does it meet my requirements?
- Does it add values?

Process Discernment

Your capability to judge the quality of the problem solving approach

- Logical inconsistency
- Lapses in attention
- Inappropriate steps
- Getting stuck on one small detail
- Getting trapped in circular reasoning

Performance Discernment

Your capability to judge AI behaviors

- Is the communication styles appropriate?
- Is the information at the right level?
- Is response to feedback appropriate?
- Is the interaction efficient?

Diligence

Taking responsibility

Rigorous, transparent, and accountable

Consider broader questions

Responsibility starts with awareness

Diligence

Creation Diligence

Your ability to be critical and intentional about which AI systems you choose to use and how you use them

- The AI systems that we use
- How we work with them
- The impacts that come from that interaction

Transparency Diligence

Your ability to be open and accurate about AI interaction with relevant stakeholders

- Who needs to know?
- How should I communicate this?
- What level of detail is needed?

Deployment Diligence

Your capability to take informed responsibility for the outputs your use or share after they've been created with AI assistance

- Verify facts
- Check for biases
- Ensure accuracy
- Check for usage rights
- And so on ...

Diligence

Ethical Frameworks

- Personal guidelines
- Organizational policies
- Professional standards
- Industry codes of conduct
- Legal and regulatory frameworks
- Staying informed is an important part of diligence

[An Example](#)

A Python Task Using 4Ds

Customer feedback analysis pipeline

- ▶ Load a CSV of customer reviews
- ▶ Clean text and missing values
- ▶ Label simple sentiment with rule-based logic
- ▶ Summarize ratings and sentiment counts
- ▶ Export a clean report

Sample data fields

review_id

customer_name

rating

comment

date

Q&A

CBPM B214

dchen@kean.edu

Tue & Thu 11:15 – 12:30 | Wed 14:00 – 16:30